

Title: University Library Data Warehouse – Security Implementation

Objectives:

- Ensure confidentiality, integrity, and availability of data.
- Protect against unauthorized access and data breaches.

Implementation Measures:

1. Authentication and Authorization

- Users authenticate via **unique credentials**.
- RBAC enforced through database views, stored procedures, and dashboard filters.
- Admin role: full CRUD; Analyst & Department Head: read-only with restrictions.

2. Data Masking

- Sensitive student identifiers (StudentID) are **masked** in dashboards.
- Personal contact data is **redacted** from reports.

3. Encryption

- **Data in transit:** TLS/HTTPS for all dashboard access.
- **Data at rest:** Database encryption enabled via MySQL AES_ENCRYPT or file-level encryption.

4. Audit Logging

- All access to dashboards and databases logged with:
 - User ID
 - Timestamp
 - Action performed (view, query, update)
- Logs reviewed weekly by the IT/security team.

5. ETL Security

- ETL scripts stored in a **restricted directory**.
- Staging tables access restricted to ETL process only.

- Error logs anonymized before sharing.

6. **Dashboard Security**

- Filters and pivot-level access based on user role.
- No direct download of sensitive raw data for restricted users.